Binary and binomial regression

Statistics III - CdL SSE

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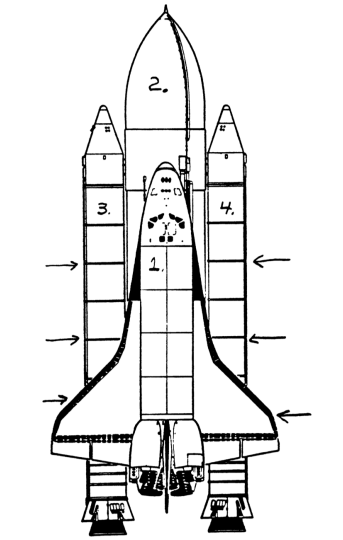


Figure 2. Space Shuttle: Orbiter, External Tank, Solid Rocket Motors, and Field Joints.

Original paper

- GLMs for **binary data** and **binomial data** are very common, whenever the response variable is dichotomous (e.g., success/failure, yes/no, dead/alive, diseased/healthy, etc.).
- This unit will cover a few additional topics related to binary and binomial regression, including:
  - grouped vs ungrouped data;
  - the choice of the link function;
  - interpretation of responses via latent "utilities";
  - a case study: the **Challenger** disaster.
- The most important aspects have already been covered in Unit B.

The content of this Unit is covered in **Chapter 3** of Salvan et al. (2020). Alternatively, see **Chapter 5** of Agresti (2015).

Notation and recap

- In a **binomial regression** model, we observe S_i **successes** out of m_i **trials**

$$S_i \overset{\text{ind}}{\sim} \text{Binomial}(m_i, \pi_i), \qquad g(\pi_i) = \eta_i = \boldsymbol{x}_i^T \beta, \qquad i = 1, \ldots, n.$$

  We model the proportions $Y_i = S_i/m_i$, whose mean is indeed π_i and $Y_i \overset{\text{ind}}{\sim} \text{ED}(\mu_i, \mu_i(1-\mu_i)/m_i)$.

- In a **binary regression** model, we observe $Y_i \in \{0, 1\}$, which is a special case of the model above with number of trials $m_i = 1$ for all i. Thus, we have

$$Y_i \overset{\text{ind}}{\sim} \text{Bernoulli}(\pi_i), \qquad g(\pi_i) = \eta_i = \boldsymbol{x}_i^T \beta, \qquad i = 1, \ldots, n.$$

In this unit, we distinguish between **two sample size measures**:

- a measure m_i for the number of Bernoulli trials that constitute a particular binomial observation;
- a measure n for the number of binomial observations.

Grouped vs ungrouped data

- If $Y_{ij} \in \{0, 1\}$ are independent **Bernoulli** random variables with success probability π_i, for $j = 1, \ldots, m_i$ and $i = 1, \ldots, n$, then

$$S_i = \sum_{j=1}^{m_i} Y_{ij} \sim \text{Binomial}(m_i, \pi_i).$$

- Thus, any binomial regression model can be **ungrouped** into a binary regression model with $N = \sum_{i=1}^{n} m_i$ observations, by replicating the covariate vector $\boldsymbol{x}_i$ m_i times, with S_i of the corresponding binary responses equal to 1 and the remaining $m_i - S_i$ equal to 0.
- On the other hand, a binary regression model can be **grouped** only if multiple subjects share the same values for explanatory variables, which is common if they are all categorical.

The **likelihood** function of these two representations **coincide** up to a proportionality constant, which means that the maximum likelihood $\hat{\beta}$ and the standard errors are identical.

However, the deviance and the residuals are different, which has implications for goodness-of-fit tests and model diagnostics. The diagnostics for binary data are typically uninformative.

Link functions I

- As discussed, the **link function** is usually set to $g(\cdot) = F^{-1}(\cdot)$ for some **continuous cumulative distribution function** $F(\cdot) : \mathbb{R} \to (0, 1)$. In other words,

$$g(\pi_i) = F^{-1}(\pi_i) = \boldsymbol{x}_i^T \beta,$$

  that is

$$\mu_i = \pi_i = F(\boldsymbol{x}_i^T \beta).$$

  The function $F(\cdot)$ is strictly increasing, differentiable, and maps the real line to the unit interval.

- **Logit link**. The canonical link is the logistic link (or logit link), that is

$$g(\pi_i) = \log\left(\frac{\pi_i}{1-\pi_i}\right) = \boldsymbol{x}_i^T \beta, \qquad \text{with inverse} \qquad \pi_i = \frac{\exp(\boldsymbol{x}_i^T \beta)}{1 + \exp(\boldsymbol{x}_i^T \beta)}.$$

  Indeed, the function $F(z) = \exp(z)/(1 + \exp(z))$ with $z \in \mathbb{R}$ is the cdf of a **logistic distribution** with mean 0 and variance $\pi^2/3$.

Link functions II

- **Probit link**. The probit link is based on the standard normal cdf $\Phi(\cdot)$, i.e.

$$g(\pi_i) = \Phi^{-1}(\pi_i) = \boldsymbol{x}_i^T\beta, \qquad \text{with inverse} \qquad \pi_i = \Phi(\boldsymbol{x}_i^T\beta).$$

  The function $\Phi(z)$ with $z \in \mathbb{R}$ is usually computed numerically and equals

$$\Phi(z) = \int_{-\infty}^{z} \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{u^2}{2}\right) du.$$

- **Complementary log-log link**. The complementary log-log (cloglog) link is

$$g(\pi_i) = \log(-\log(1-\pi_i)) = \boldsymbol{x}_i^T\beta, \qquad \text{with inverse} \qquad \pi_i = 1 - \exp(-\exp(\boldsymbol{x}_i^T\beta)).$$

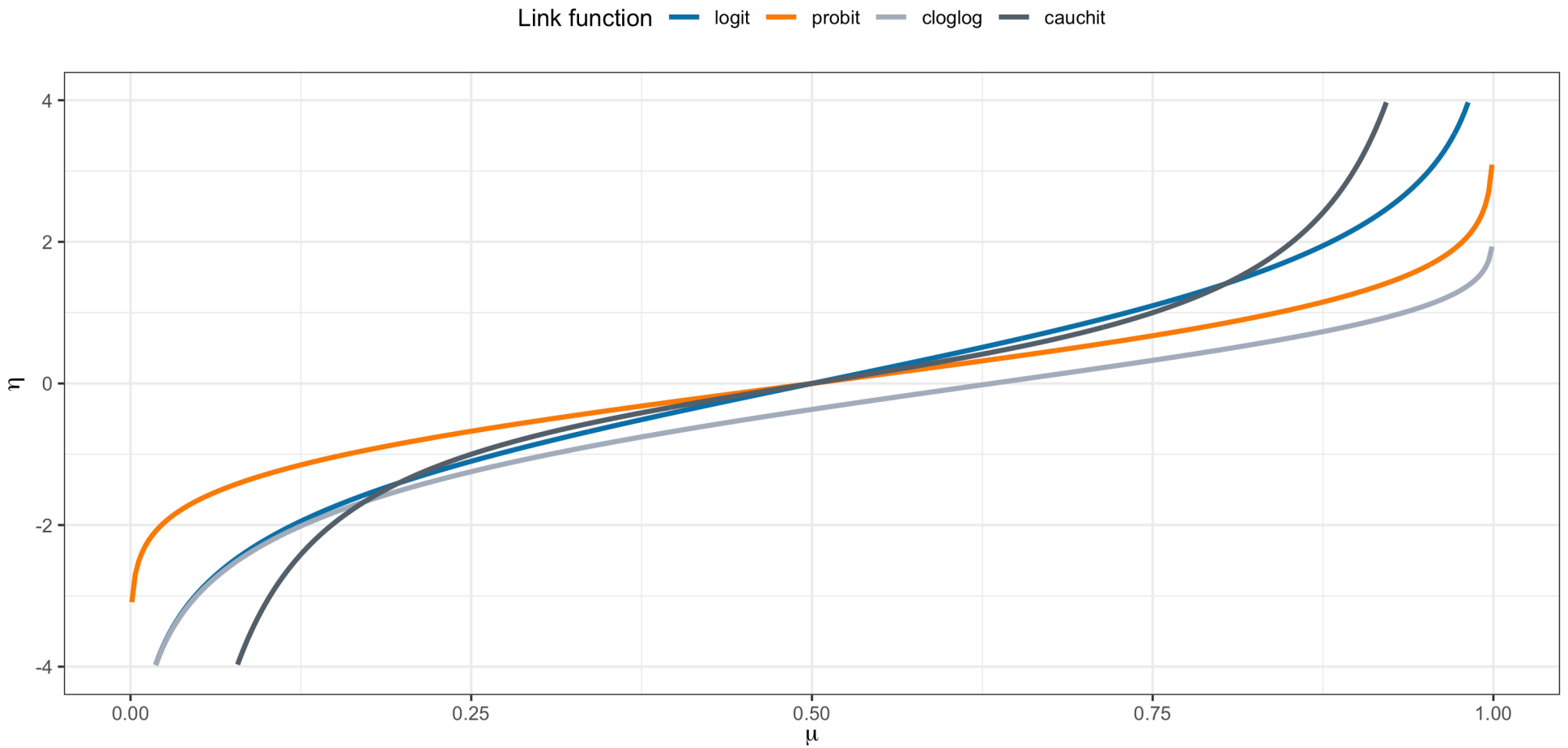
  The distribution function $F(z) = 1 - \exp(-\exp(z))$ with $z \in \mathbb{R}$ is the cdf of a Gumbel distribution (for minima). This link function is **asymmetric**.

- **Cauchy link**. The Cauchy link is

$$g(\pi_i) = \tan(\pi(\pi_i - 1/2)), \qquad \text{with inverse} \qquad \pi_i = \frac{1}{2} + \frac{\arctan(\boldsymbol{x}_i^T\beta)}{\pi}.$$

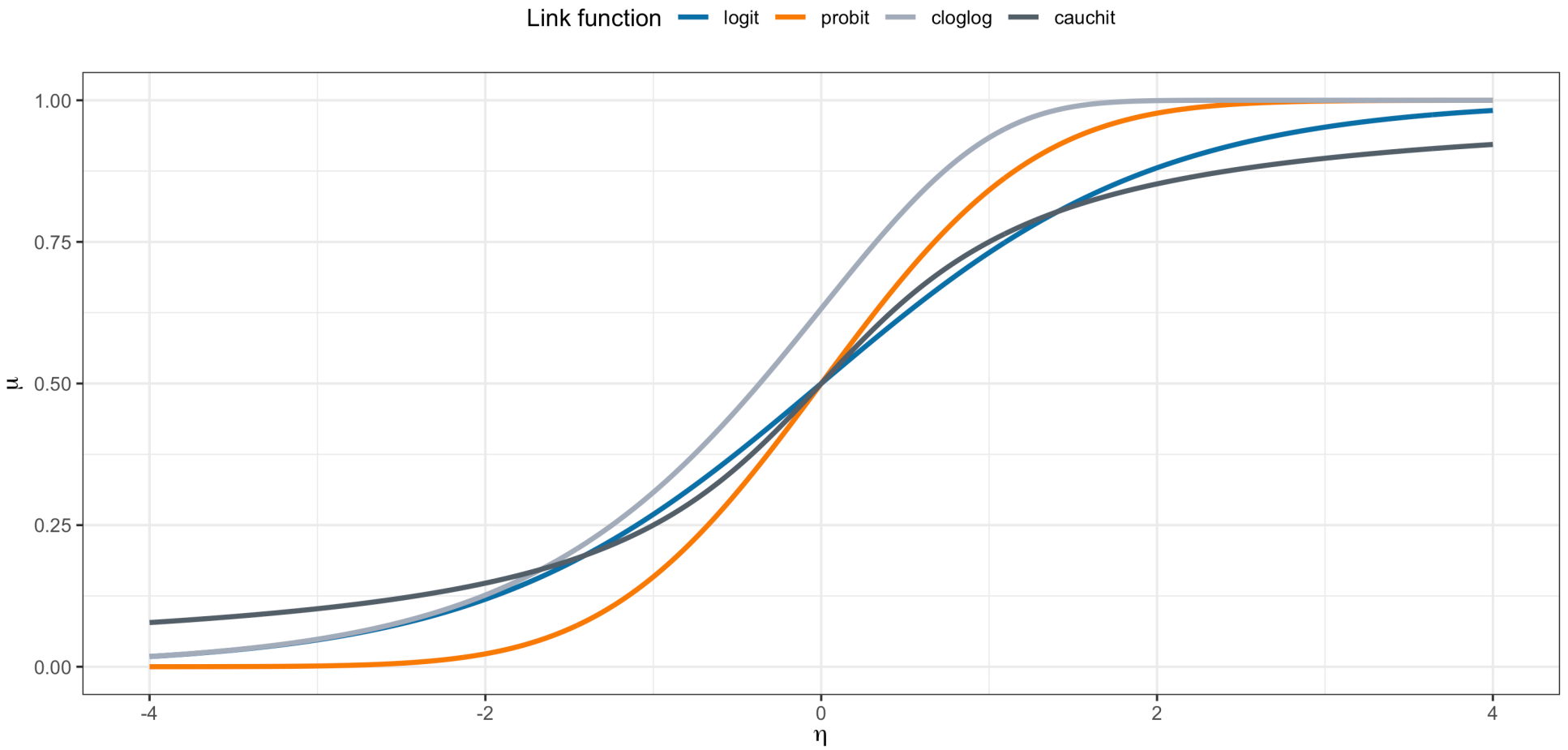
  $F(z) = 1/2 + \arctan(z)/\pi$ with $z \in \mathbb{R}$ is the cdf of a **standard Cauchy distribution**.

Link functions III



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Link functions IV



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Latent variable threshold models I

- A latent variable threshold model is a useful way to interpret binary regression models and to choose their link function $g(\cdot)$. Let Y_i^* be a **latent** and **continuous** random variable such that

$$Y_i^* = \boldsymbol{x}_i^T \beta + \epsilon_i, \qquad \epsilon_i \overset{\text{iid}}{\sim} F.$$

- A variable is called latent because we do not observe it directly. Instead, we observe only a **binary variable** encoding whether it exceeds a certain threshold, i.e.

$$Y_i = \mathbb{I}(Y_i^* > \tau),$$

  where $\mathbb{I}(\cdot)$ is the indicator function and τ is a threshold.

- By construction, we have $Y_i \sim \text{Bernoulli}(\pi_i)$, with

$$\pi_i = \mathbb{P}(Y_i = 1) = \mathbb{P}(Y_i^* > \tau) = \mathbb{P}(\boldsymbol{x}_i^T \beta + \epsilon_i > \tau) = 1 - \mathbb{P}(\epsilon_i \leq \tau - \boldsymbol{x}_i^T \beta) = 1 - F(\tau - \boldsymbol{x}_i^T \beta).$$

- The data contain **no information** about τ, so we can set $\tau = 0$ without loss of generality. Equivalently, the value of τ is absorbed into the intercept term.
- Likewise, an equivalent model results if we multiply all parameters by any positive constant, so we can take F to have a **standard form** with fixed variance, such as the standard normal cdf.

Latent variable threshold models II

- For most models F corresponds to a cdf that is symmetric around 0, so $F(z) = 1 - F(-z)$. Thus, we obtain

$$\pi_i = 1 - F(-\boldsymbol{x}_i^T\beta) = F(\boldsymbol{x}_i^T\beta), \qquad \text{and} \qquad F^{-1}(\pi_i) = g(\pi_i) = \boldsymbol{x}_i^T\beta.$$

  That is, models for binary data naturally take the **link function** to be the **inverse of the standard cdf** for a family of continuous distributions for a latent variable.

This dichotomization often corresponds to a real process — for instance, in medical diagnosis, preterm birth can be seen as a dichotomization of gestational age at delivery.

However, there are cases where the existence of a latent quantity is more questionable, such as Y_i^* corresponding to a notion of "ability" and Y_i to "passing an exam".

Logistic regression

- The **likelihood equations** of a binomial regression model are fairly simple:

$$\sum_{i=1}^{n} \frac{m_i(y_i - \pi_i)}{\pi_i(1-\pi_i)} x_{ir} f(\eta_i) = 0, \qquad r = 1, \ldots, p,$$

  where $f(z)$ is the density associated with $F(z)$, i.e. its derivative, and the **link** is $g(\cdot) = F^{-1}(\cdot)$.

- In logistic regression $f(\eta_i) = F(\eta_i)(1 - F(\eta_i)) = \pi_i(1-\pi_i)$, therefore

$$\sum_{i=1}^{n} m_i(y_i - \pi_i) x_{ir} = 0, \qquad r = 1, \ldots, p.$$

  The solution has a nice interpretation as a **method of moments** estimator, in that

$$\sum_{i=1}^{n} s_i x_{ir} = \sum_{i=1}^{n} \mathbb{E}(S_i) x_{ir}, \qquad r = 1, \ldots, p.$$

  Moreover, the estimated covariance matrix of $\hat{\beta}$ has a simple form:

$$\widehat{\mathrm{var}}(\hat{\beta}) = (\boldsymbol{X}^T \hat{\boldsymbol{W}} \boldsymbol{X})^{-1} = (\boldsymbol{X}^T \mathrm{diag}[m_1\hat{\pi}_1(1-\hat{\pi}_1), \ldots, m_n\hat{\pi}_n(1-\hat{\pi}_n)]\boldsymbol{X})^{-1}.$$

Parameter interpretation

- We wish to **compare two estimated probabilities**, corresponding to two different covariate vectors $\boldsymbol{x} = (x_1, \ldots, x_p)$ and $\boldsymbol{x}' = (x'_1, \ldots, x'_p)$, namely

$$\pi(\boldsymbol{x}) = F(\boldsymbol{x}^T\beta) \qquad \text{and} \qquad \pi(\boldsymbol{x}') = F((\boldsymbol{x}')^T\beta).$$

  This is useful to understand the effect of changing covariates on the response probability.
- There are several ways to compare them, listed below, each with its advantages and disadvantages:
  - The **risk difference**, namely the difference

$$(\text{risk difference}) = \pi(\boldsymbol{x}') - \pi(\boldsymbol{x}).$$

  - The **relative risk** is the fraction

$$(\text{relative risk}) = \frac{\pi(\boldsymbol{x}')}{\pi(\boldsymbol{x})}.$$

- Both these indicators are quite interpretable but depend on the specific values of $\boldsymbol{x}$ and $\boldsymbol{x}'$, which makes it difficult to summarize the effect of a given covariate in a single number.

Odds ratio

- The **odds** is another way of summarizing probabilities, familiar to those who gamble:

$$\text{odds}(\boldsymbol{x}) = \frac{\pi(\boldsymbol{x})}{1-\pi(\boldsymbol{x})}.$$

  The odds are non-negative, with $\text{odds}(\boldsymbol{x}) > 1$ when a **success is more likely than a failure**.

- When $\pi(\boldsymbol{x}) = 0.75$ then $\text{odds}(\boldsymbol{x}) = 0.75/0.25 = 3$; a success is three times as likely as a failure. That is, we bet 3 to get 1 if we win. If instead $\text{odds}(\boldsymbol{x}) = 1/3$, we bet 1 to get 3 if we win.

- The **odds ratio** is another popular risk measure for comparing the two probabilities:

$$(\text{odds ratio}) = \frac{\text{odds}(\boldsymbol{x}')}{\text{odds}(\boldsymbol{x})} = \frac{\pi(\boldsymbol{x}')}{\pi(\boldsymbol{x})} \cdot \frac{1-\pi(\boldsymbol{x})}{1-\pi(\boldsymbol{x}')}.$$

- Let $\boldsymbol{x} = (x_1, \ldots, x_j, \ldots, x_p)$ and $\boldsymbol{x}' = (x_1, \ldots, x_j + c, \ldots, x_p)$, i.e. we compare two situations in which the jth covariate is increased by a fixed amount c, then in **logistic regression**:

$$(\text{odds ratio}) = \exp(c\beta_j),$$

  which is a **constant number** that does not depend on $\boldsymbol{x}$. This is not true e.g. in probit.

Odds ratio and logistic regression

- Let $\boldsymbol{x}$ be a generic covariate vector and consider its odds under a **logistic regression** model:

$$\text{odds}(\boldsymbol{x}) = \frac{\pi(\boldsymbol{x})}{1-\pi(\boldsymbol{x})} = \frac{\exp(\boldsymbol{x}^T\beta)}{1+\exp(\boldsymbol{x}^T\beta)} \cdot \left(\frac{1}{1+\exp(\boldsymbol{x}^T\beta)}\right)^{-1} = \exp(\boldsymbol{x}^T\beta).$$

  Thus, the odds at any covariate value are simply the exponential of the linear predictor $\boldsymbol{x}^T\beta$.

- Consequently, in general the **odds ratio** is

$$(\text{odds ratio}) = \frac{\text{odds}(\boldsymbol{x}')}{\text{odds}(\boldsymbol{x})} = \exp\{(\boldsymbol{x}'-\boldsymbol{x})^T\beta\}.$$

- If the only change is in the jth covariate, from x_j to $x_j + c$, then the odds ratio becomes

$$(\text{odds ratio}) = \exp\{(\boldsymbol{x}'-\boldsymbol{x})^T\beta\} = \exp(c\beta_j).$$

  In particular, $\exp(\beta_j)$ represents the odds ratio after a **unit increase** of the jth covariate.

Residuals and diagnostics for binary data

- In presence of **binary data**, namely when $m_i = 1$ and $Y_i \sim \text{Bernoulli}(\pi_i)$, many of the diagnostic tools are **degenerate** and uninformative.
- For example, the `fitted` vs `residuals` plot, namely the scatter plot of the points $(\hat{\pi}_i, y_i - \hat{\pi}_i)$, simply shows two parallel lines with slope -1, which is uninformative.
- As an extreme case, consider the **null model** in which $\pi_1 = \cdots = \pi_n$, then the X^2 statistic is

$$X^2 = \sum_{i=1}^{n} \frac{(y_i - \hat{\pi})^2}{\hat{\pi}(1 - \hat{\pi})} = n,$$

  whatever the observed data, since $\hat{\pi} = \bar{y}$ and therefore $\sum_{i=1}^{n}(y_i - \bar{y})^2 = n\bar{y}(1 - \bar{y})$.
- A common solution is to **group the data** according to some criterion, although this introduces approximations and some arbitrariness (e.g. the Hosmer–Lemeshow test).
- In terms of prediction, there exist various indices and techniques that you will cover in Data Mining. The simplest index is the **accuracy**, i.e. the fraction of correct predictions.

Overdispersion

- In binomial regression the main **assumption** is that $S_i \sim \text{Binomial}(m_i, \pi_i)$, implying that

$$\text{var}(Y_i) = \frac{\pi_i(1-\pi_i)}{m_i},$$

  where implicitly we have set $\phi = 1$.

- However, from the analysis of the residuals or by computing the X^2 statistic we may realize that the data present **overdispersion**, namely a **more appropriate model** is such that

$$\text{var}(Y_i) = \phi\frac{\pi_i(1-\pi_i)}{m_i},$$

  with $\phi > 1$. This implies that the binomial regression model is **misspecified**.

- Overdispersion can only occur with **grouped data**: when $m_i = 1$, any distribution on $\{0, 1\}$ has variance $\pi_i(1-\pi_i)$, so necessarily $\phi = 1$.

Case study: the `shuttle` data

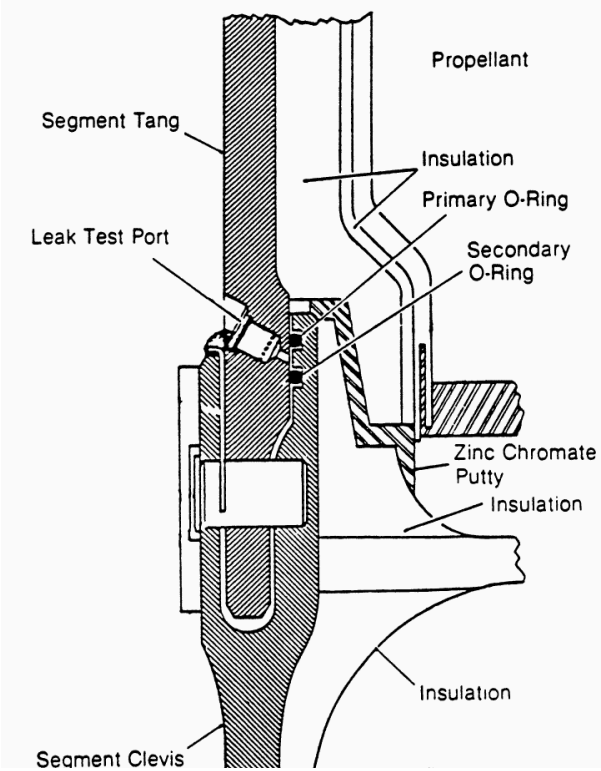
The Challenger disaster



- The space shuttle **Challenger** broke apart 73 seconds after its launch on **28 January 1986**. All seven crew members died.
- The Presidential Commission found that the cause was a **gas leak** through a joint of the right solid rocket motor.
- The joint was sealed by rubber **O-rings**, which do not seal properly at low temperatures.
- The forecast for the launch was **31°F** (about -0.5°C).

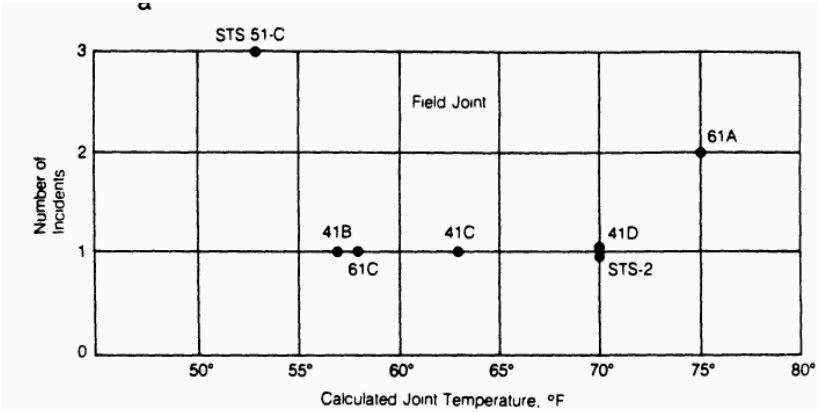
- Netflix docuseries: *Challenger: The Final Flight* (2020)
- Dalal et al. (1989), *Risk analysis of the space shuttle: pre-Challenger prediction of failure*, JASA.

O-rings and thermal distress



- The two solid rocket motors have **six field joints** in total. Each joint is sealed by a **primary** and a secondary O-ring.
- A primary O-ring experiences **thermal distress** if:
  - hot gas burns part of the ring (**erosion**);
  - hot gas leaks past the ring before it seals (**blowby**).
- Bench tests had shown that O-rings **lose their resilience** at **low temperatures**.
- During a public hearing, Richard Feynman showed this by dipping a piece of O-ring in a glass of ice water.

The night before the launch



- On 27 January 1986, engineers of **Morton Thiokol**, the manufacturer of the motors, and NASA discussed the launch in a teleconference.
- The discussion focused on the plot on the left:
  - it displays the **7 flights** with at least one O-ring incident;
  - the **16 flights** with no incidents were left out.
- Some engineers asked to postpone the launch, but Thiokol concluded that "*temperature data [are] not conclusive*".
- The launch was approved.

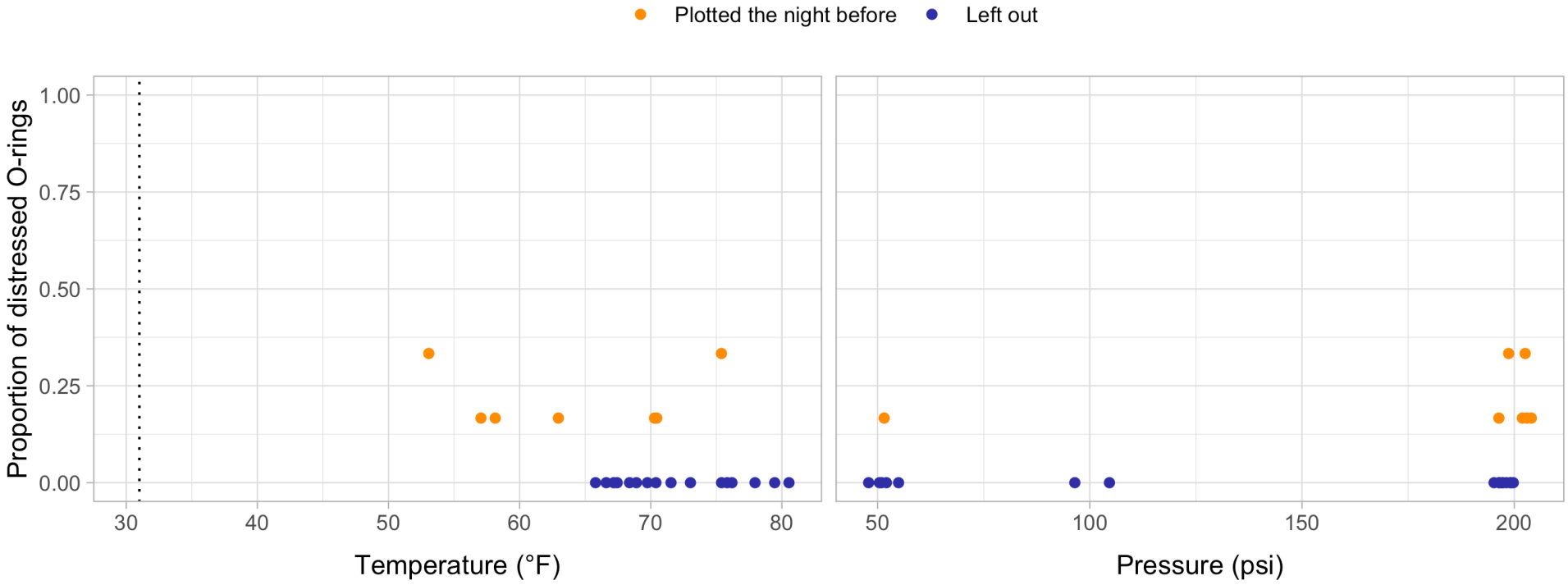
The `shuttle` dataset

- The `shuttle` dataset of the `SMPracticals` library contains the data of the **23 flights** before the accident.
  - `r`: number of primary field-joint O-rings with thermal distress, out of `m` $= 6$;
  - `temperature`: calculated joint temperature at launch (°F);
  - `pressure`: pressure of the leak-check test performed before each launch (psi).

```
  m r temperature pressure
1 6 0          66       50
2 6 1          70       50
3 6 0          69       50
4 6 0          68       50
5 6 0          67       50
6 6 0          72       50
```

- There were 24 launches before the accident, but the motors of one flight were lost at sea.
- The leak-check pressure was **increased** over time: 50 psi in the first six flights, 100 psi in the next two, then 200 psi.

Exploratory analysis



- As **temperature** decreases, the proportion of distressed O-rings **increases**. There is less pattern for **pressure**.
- All the **16 flights left out** the night before were launched at 66°F or above.
- The dotted line is the 31°F forecast, 22°F below the coldest previous launch (53°F).

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Modelling the `shuttle` data

- Let S_i be the number of distressed O-rings in the ith flight. If, within each flight, the six O-rings fail **independently** and with the **same probability** π_i, then

$$S_i \overset{\text{ind}}{\sim} \text{Binomial}(6, \pi_i), \qquad i = 1, \ldots, 23.$$

- We consider a **logistic regression** model, with x_{i1} the temperature and x_{i2} the pressure:

$$\text{logit}(\pi_i) = \log\left(\frac{\pi_i}{1-\pi_i}\right) = \beta_1 + \beta_2 x_{i1} + \beta_3 x_{i2}.$$

- Our main goal is the **probability of thermal distress at 31°F** which, if pressure has no effect ($\beta_3 = 0$), is

$$\psi = \frac{\exp(\beta_1 + 31\beta_2)}{1 + \exp(\beta_1 + 31\beta_2)}.$$

The full model

```
Call:
glm(formula = cbind(r, m - r) ~ temperature + pressure, family = binomial,
    data = shuttle)

Coefficients:
             Estimate Std. Error z value Pr(>|z|)
(Intercept)  2.520195   3.486784   0.723   0.4698
temperature -0.098297   0.044890  -2.190   0.0285 *
pressure     0.008484   0.007677   1.105   0.2691
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 24.230  on 22  degrees of freedom
Residual deviance: 16.546  on 20  degrees of freedom
AIC: 36.106
```

- Given temperature, the Wald test for $H_0 : \beta_3 = 0$ gives $W_e = 1.105^2$ (p-value 0.269). Any **pressure** effect is **slight** or absent.

The temperature model

```
Call:
glm(formula = cbind(r, m - r) ~ temperature, family = binomial,
    data = shuttle)

Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept)  5.08498    3.05247   1.666   0.0957 .
temperature -0.11560    0.04702  -2.458   0.0140 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 24.230  on 22  degrees of freedom
Residual deviance: 18.086  on 21  degrees of freedom
AIC: 35.647

Number of Fisher Scoring iterations: 5
```

- Without pressure, the Wald test for $H_0 : \beta_2 = 0$ gives $W_e = 2.458^2$ (p-value 0.014). There is evidence of a **temperature** effect.

Interpretation of the coefficients

- The **estimated probability** that an O-ring experiences thermal distress at temperature x is

$$\hat{\pi}(x) = \frac{\exp(5.085 - 0.116\,x)}{1 + \exp(5.085 - 0.116\,x)}.$$

- The intercept $\hat{\beta}_1$ is **not of interest**:
  - it is the logit of the probability of distress at 0°F, i.e. $\hat{\pi}(0) = 0.994$;
  - 0°F is far outside the range of the data.
- An **increase** of 1°F multiplies the **odds** of distress by $\exp(\hat{\beta}_2) = 0.891$, i.e. the odds decrease by 10.9%, whatever the starting temperature.
- A **decrease** of 10°F, from x to $x - 10$, gives the **odds ratio**

$$\frac{\widehat{\text{odds}}(x-10)}{\widehat{\text{odds}}(x)} = \frac{\exp\{\hat{\beta}_1 + \hat{\beta}_2(x-10)\}}{\exp(\hat{\beta}_1 + \hat{\beta}_2 x)} = \exp(-10\hat{\beta}_2) = 3.177,$$

that is, the odds increase by $100 \times (3.177 - 1) = 217.7\%$, whatever the starting temperature.

Probabilities, odds and risks

- Estimated probabilities $\hat{\pi}(x)$ and odds $\widehat{\text{odds}}(x) = \hat{\pi}(x)/\{1 - \hat{\pi}(x)\}$ of thermal distress:

	$x = 70$°F	$x = 60$°F	$x = 50$°F	$x = 31$°F
Probability $\hat{\pi}(x)$	0.047	0.136	0.333	0.818
Odds $\widehat{\text{odds}}(x)$	0.049	0.157	0.499	4.488

- For a **decrease of 10°F**, from x to $x - 10$, we compare $\hat{\pi}(x - 10)$ with $\hat{\pi}(x)$:

Temperature (°F)	Risk difference	Relative risk	Odds ratio
70 → 60	0.089	2.882	3.177
60 → 50	0.197	2.452	3.177

- The **risk difference** and the **relative risk** depend on the starting temperature x, whereas the **odds ratio** is always $\exp(-10\hat{\beta}_2) = 3.177$.

The odds ratio is a variation of the **odds**, **not** of the **probabilities**. For instance, from 60°F to 50°F the probability of distress is multiplied by 2.452, not by 3.177.

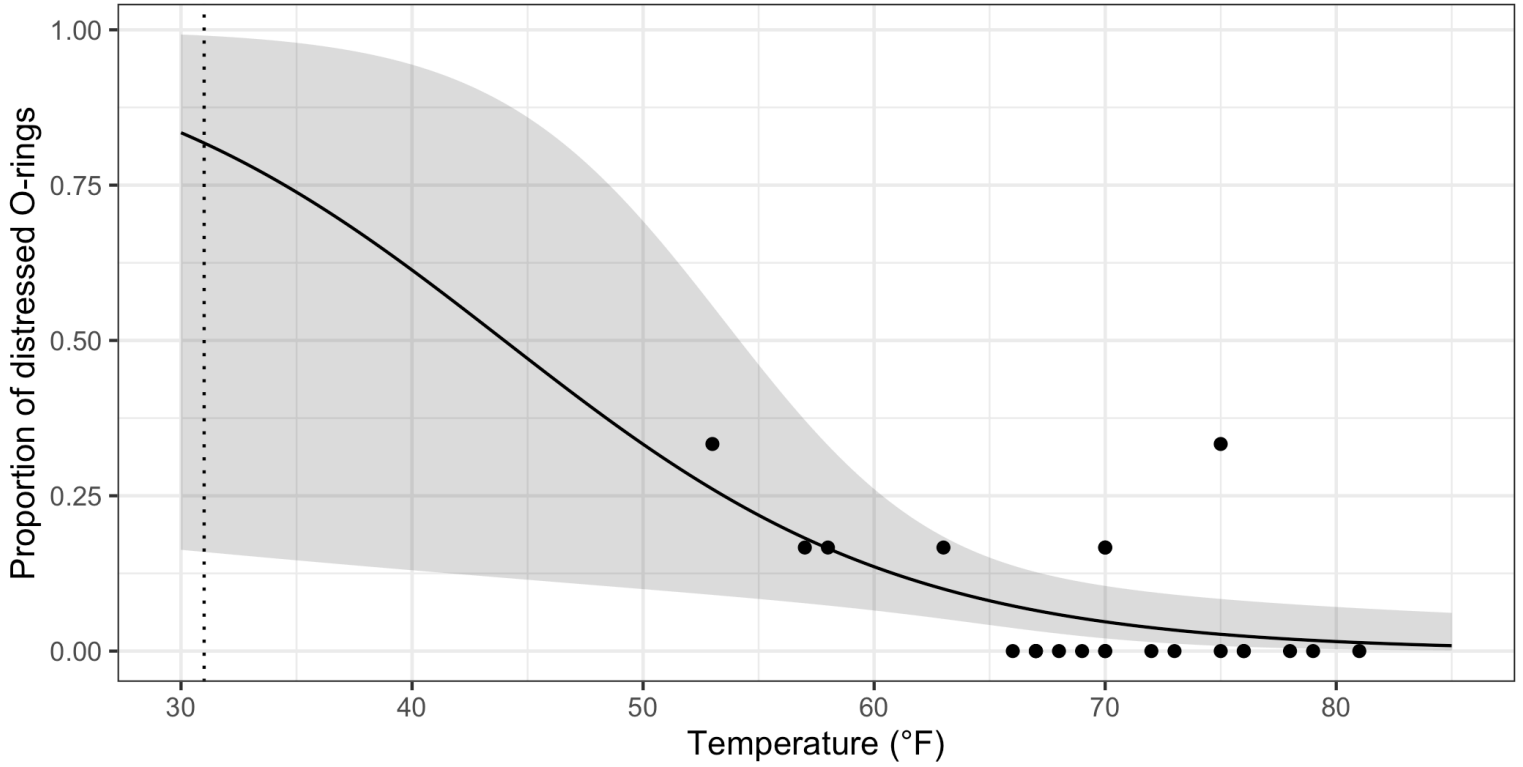
Probability of distress at 31°F

- The estimated probability of distress at 31°F is $\hat{\psi} = 0.818$. We expect $6\hat{\psi} = 4.9$, i.e. **four to five**, distressed O-rings out of six.

	Lower	Upper
Wald, delta method	0.346	1.289
Wald, logit scale	0.160	0.991

- The Wald interval for ψ based on the **delta method** contains values **greater than one**, so it is inappropriate.
- The Wald interval computed on the **logit scale** and then transformed, as in Unit B, always lies in $(0, 1)$.

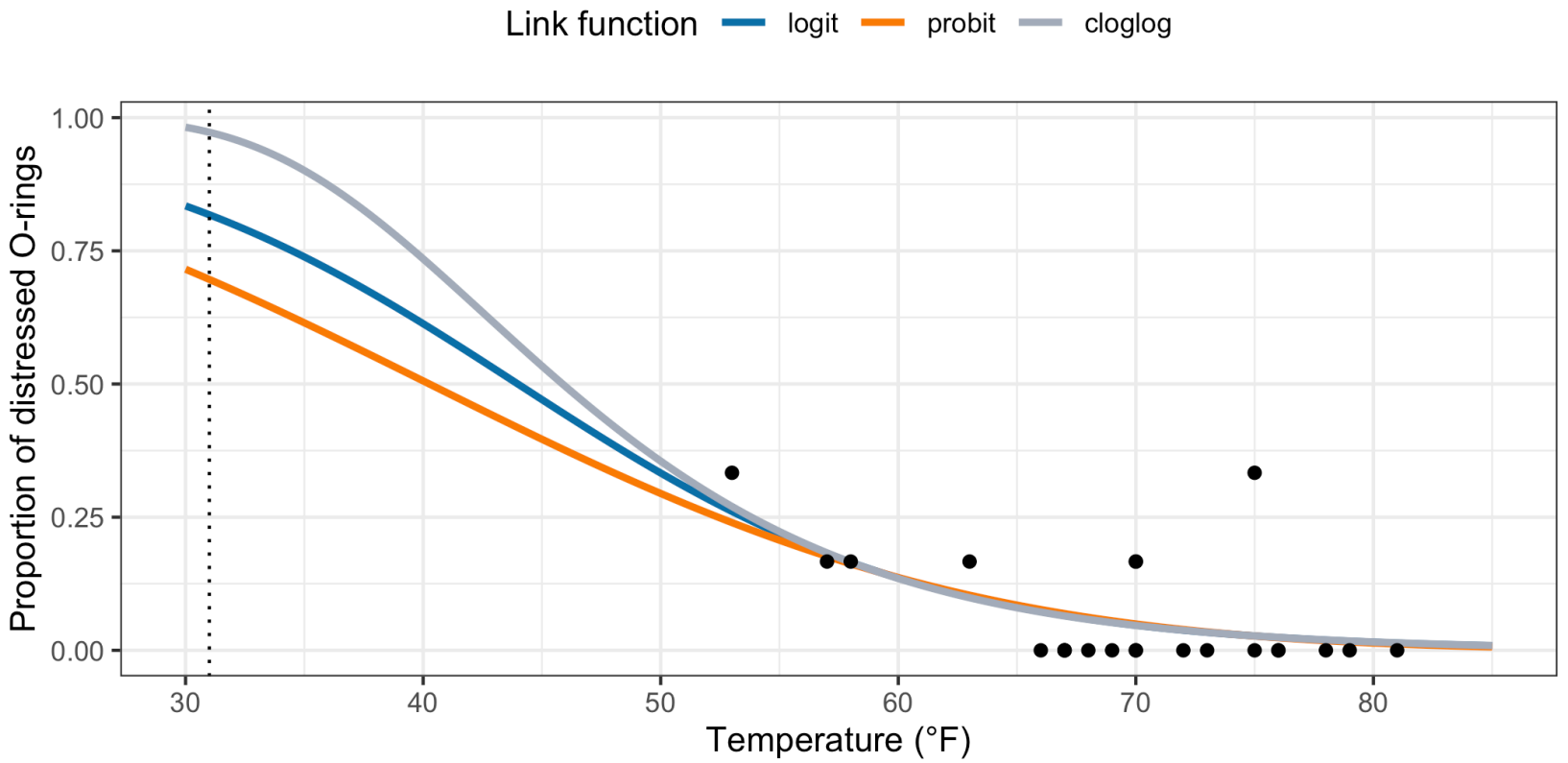
Estimated probability of distress



- Estimated probability of distress, with pointwise 95% confidence intervals computed on the logit scale.
- The prediction at 31°F is an **extrapolation** far outside the data, but there would have been little alternative.

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Choice of the link function



- The deviances are 18.09 (logit), 18.31 (probit) and 18.01 (cloglog): the three fits are almost identical in the range of the data.
- However, the estimated probabilities at 31°F are **different**: 0.82, 0.70 and 0.97, respectively.

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Conclusions

- **Goodness of fit**: the deviance is 18.09 on 21 degrees of freedom and $X^2 = 29.98$.
  - The counts are small and the χ^2 approximation is rough, but there is no clear evidence of lack of fit.
- The **temperature effect** is robust: logit, probit and cloglog links fit the data equally well and agree on its sign.
- At 31°F we are **extrapolating**, 22°F below any previous launch: the 95% confidence interval for ψ is (0.16, 0.99).
- Yet all models point to a **high probability** of distress at 31°F, and no data supported the safety of such a launch.

The prediction is uncertain, but given the **catastrophic consequences** of a failure, caution would have required to **postpone the launch**.

Epilogue



The crew of the Challenger. Source: NASA.

- The Rogers Commission: "*a careful analysis of the flight history of O-ring performance would have revealed the correlation of O-ring damage in low temperature*".
- Building on this model, Dalal et al. (1989) estimated a probability of **catastrophic failure** of at least 0.13 at 31°F, against 0.02 at 60°F.
- After the accident, NASA started to perform probabilistic risk assessments. Crewed flights resumed only in September 1988.

The lesson: **never discard the zeros**. The flights with no incidents are the ones showing that warm launches were safe.

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